

SAEHUN SEAN OH

Senior Software Engineer · Android, Streaming Clients & Data Platform

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SUMMARY

Senior engineer with 11 years in consumer video, spanning Android, Roku, and Chromecast clients and the analytics platform they feed. Currently own a dbt / Airflow / BigQuery platform after a decade of client engineering, which means I design the events, ship the surfaces that emit them, and answer for the numbers they produce. Twice self-initiated and shipped on a platform I had never used.

SKILLS

Languages: Kotlin, TypeScript, Python, SQL, Java, BrighterScript

Android: Jetpack Compose, MVVM, Hilt, Coroutines & Flow, Room, DataStore, ExoPlayer, Retrofit / OkHttp, JUnit / MockK, Firebase, Google Play Billing, Gradle

Connected TV & Web: Roku (SceneGraph, BrighterScript), Chromecast (CAF Web Receiver, TypeScript, Vercel), Android TV / Fire TV / webOS / Tizen

Data & Backend: dbt, Apache Airflow, Google BigQuery, Apache Kafka, Google Analytics 4, AWS (Lambda, S3), REST API design, event schema design

Streaming & Ads: HLS and manifest parsing, CSAI / SSAI, Mux quality-of-experience analytics, PubMatic OpenWrap header bidding, ad mediation, OneTrust (CCPA consent)

Tooling & AI: GitHub Actions CI/CD, automated UI testing, Git submodule shared libraries, Claude Code (agent skills, subagent workflows, CLAUDE.md), ChatGPT

Languages Spoken: English, Korean (bilingual)

EXPERIENCE

ODK Media — Streaming / OTT · OnDemandKorea & Amasian

2020 – Present

Sr. Data & Platform Engineer · Sr. Roku Engineer · Project Coordinator, CTV Apps · Sr. Android Engineer

Data & Analytics Platform

2026 – Present

- Own the analytics platform (dbt / Airflow / BigQuery) and set the data-modeling standards for a new pipeline replacing a legacy Kafka / Luigi chain and its layered ad-hoc scheduled queries.
- Built a daily 3-way validation reconciling Kafka, GA4, and Mux, cutting the daily scan from 186 GB to 1.5 GB (~99%) by rebuilding its source as a partitioned dbt model.
- Automated a manual partner settlement process: data transfer into S3 with validation, scheduled queries migrated to service-account ownership with backfill, and duplicated settlement rows corrected.
- Built an internal Slack tool (Bolt on AWS Lambda) that lets non-engineers investigate and acknowledge data-quality issues through slash commands.
- Own audience-data strategy for addressable advertising: content-affinity and language pools, audience sizing, and the ad-impression analytics feed behind an internal BI tool, with identity federation research (ID5, UID 2.0, LiveRamp).

Client Engineering — Android, Roku, Chromecast

2020 – 2026

- Built the first-generation OnDemandKorea Android app, then led the ground-up v2 rewrite in Kotlin and Jetpack Compose on an MVVM architecture with Hilt, including a major player refactor and a rewrite of the API-layer repositories.
- Sustained 99% crash-free sessions across 2M+ monthly view hours using Firebase and Mux monitoring routed into Slack alerting.
- Implemented local persistence with Room and DataStore, covered ViewModel and API-layer functions with JUnit and MockK tests, and eliminated unnecessary Compose recompositions through profiling.
- Sole Roku engineer for both apps (BrighterScript / SceneGraph): deep linking across all content types, multiple platform certifications, EPG promo banners, and paginated live recommendations.
- Wrote a custom subtitle and closed-caption renderer for Roku, which cannot display non-Latin characters natively, parsing the HLS manifest to locate and process subtitle sources.
- Self-initiated the Chromecast effort when nobody was working on it: taught myself Google's CAF Web Receiver with no prior web background, shipped two production TypeScript receivers deployed on Vercel with SSAI behavior and low-memory device optimization, and wrote the sender-side Cast integration in both Android apps.

- Diagnosed a circular-reference memory leak inside a third-party player analytics SDK, proposed the fix, and worked with the vendor's engineer to get it released.
- Shipped engagement features including a OneSignal notification system driving 18K+ app visits per month and Google Play and Roku Billing with backend purchase validation.

Instrumentation, Ads & Privacy

- Wired player state into the Mux analytics layer across platforms and standardized event metadata so playback could be compared app to app rather than measured differently on each.
- Parsed HLS manifests on Roku to drive ad analytics, logging impressions, first view, quartile completions, and pod boundaries into Mux, Roku Analytics, and internal player events.
- Integrated third-party ad SDKs including PubMatic OpenWrap header bidding, and the OneTrust SDK for CCPA compliance with a layered consent gate passing correct us_privacy strings downstream.

Tooling, AI & Team Practice

- Extracted the Android API layer into a shared submodule consumed by both apps, and built the equivalent versioned Roku data library with automated sync and linting tooling.
- Automated the build and release pipeline with GitHub Actions and built automated UI testing driven through Roku's debugging protocol, replacing manual release-candidate handovers.
- Earliest advocate for AI-assisted development across the US engineering teams; shipped production Roku code with AI assistance for over a year before the tooling was reliable, co-authored the shared CLAUDE.md configurations and reusable Claude Code skills now used beyond my team, and set guardrails hard-gating sensitive PII work behind human approval.
- Point of contact between the US office and an offshore team in Vietnam building smart-TV apps across five platforms (LG webOS, Samsung Tizen, Vizio, Fire TV, Android TV), managing requirements, scope, and releases across time zones.

Smule

2016 – 2020

Sr. Android Application Developer

- Built social graph features on a consumer social product with ~13M installs: adding and removing friends, friend recommendation, and group membership, and owned the family product letting users belong to multiple groups.
- Shipped the engagement surface on top of it, including the activity feed, comments, likes, direct messaging, and live streaming, plus third-party sharing SDK integrations. Acted as a product counterpart, giving feedback on specs and running A/B tests.
- Owned an aggregated ad SDK adapter on the ad-supported team, mediating banner and native inventory across four-plus vendors to maximize fill rate.
- Led the codebase transition from Java to Kotlin, set the team's coding conventions, and maintained a 99% crash-free rate through Firebase Crashlytics monitoring.

DiscreteUnits — Android Application Developer (Contract)

2016

- Built a restaurant ordering app (menu browsing, tab open/close, in-app ordering); integrated Parse, Facebook, Clover, and Stripe for credentials, catalog, and payments.

MaXentric — Android Application Developer

2015 – 2016

- Built the MaXJacket front-end app replacing Motorola radio hardware with VoIP alongside P25; used GPS / Google Maps and Android sensors; raised crash-free users by 40%.

PROJECTS

Timely Quest — gamified focus timer built at UC San Diego, turning study sessions into quest progression.

saehunseanoh.me — personal site and portfolio; separately built a small GraphQL project to learn the stack end to end.

EDUCATION

B.S. Computer Science · University of California, San Diego